Yakeen Plus (2025)

Short Practice Test - 07

DURATION: 60 Minutes | **DATE: 30/10/2024** | **M. MARKS: 192**

Topics Covered

Physics:	Current Electricity, Moving Charges and Magnetism, Magnetism and Matter, Electromagnetic Induction
Chemistry:	Aldehydes, Ketones and Carboxylic Acids, Amines, Haloalkanes and Haloarenes, Alcohols, Ethers and Phenols
Biology:	**(Botany) :** Photosynthesis in Higher Plants, Plant Growth and Development, Sexual Reproduction in Flowering Plant
	(Zoology) : Animal Kingdom, Biomolecules, Human Reproduction

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **60 minutes** duration.
3. The test booklet consists of **48** questions. The maximum marks are **192**.
4. All questions are compulsory.
5. There is only one correct response for each question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : ____________________

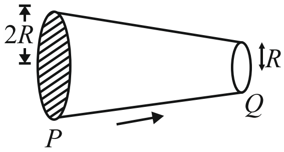
Roll Number : ____________________

OMR Bar Code Number : ____________________

Candidate's Signature : ____________________ **Invigilator's Signature** ____________________

SECTION-(I) PHYSICS

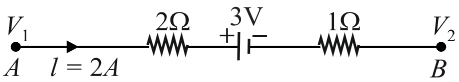
1. Electrical current is passing through a solid conductor PQ from P to Q. The electric current densities at P and Q are in the ratio.



(1) 1 : 2 (2) 2 : 1
(3) 1 : 4 (4) 4 : 1

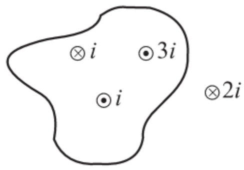
2. Two electric bulbs whose resistance are in the ratio of 1 : 2, are connected in parallel to a constant voltage source. The power dissipated in them has the ratio.
(1) 2 : 1
(2) 1 : 1
(3) 1 : 4
(4) 1 : 2

3. The potential difference ($V_A - V_B$) between the points A and B in the figure is:



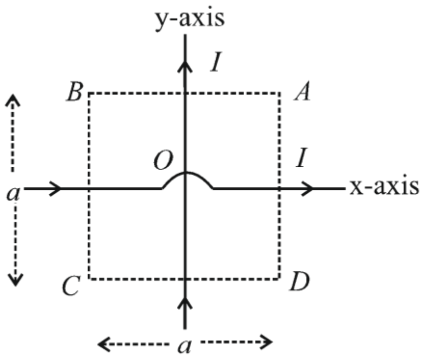
(1) –5V (2) +3V
(3) +6V (4) +9V

4. The value of $\oint \vec{B} \cdot d\vec{\ell}$ for the loop as shown in the figure will be:



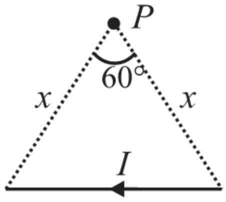
(1) $2\mu_0 i$ (2) $3\mu_0 i$
(3) $\mu_0 i$ (4) $5\mu_0 i$

5. Two infinite length wires carry equal current and placed along x and y axis respectively. At which points the resultant magnetic field is zero?



(1) A, B (2) B, D
(3) A, C (4) C, D

6. A straight wire of finite length carrying current I subtends an angle of 60° at point P as shown. The magnetic field at P is:



(1) $\frac{\mu_0 I}{2\sqrt{3}\pi x}$ (2) $\frac{\mu_0 I}{2\pi x}$
(3) $\frac{\sqrt{3}\mu_0 I}{2\pi x}$ (4) $\frac{\mu_0 I}{3\sqrt{3}\pi x}$

7. Flux ϕ (in weber) in a closed circuit of resistance 10 ohm varies with time t (in second) according to the equation $\phi = 6t^2 - 5t + 1$. What is the magnitude of the induced current at $t = 0.25\ s$?
(1) 1.2 A (2) 0.2 A
(3) 0.6 A (4) 0.8 A

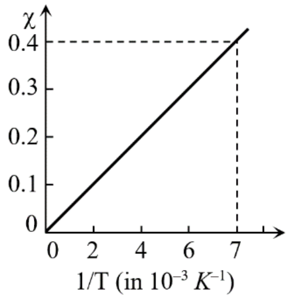
8. According to Faraday's law of electromagnetic induction:
(1) electric field is produced by time varying magnetic flux.
(2) magnetic field is produced by time varying electric flux.
(3) magnetic field is associated with a moving charge.
(4) None of these

9. The magnetic flux through a circuit of resistance R changes by an amount $\Delta\phi$ in a time Δt. Then the total quantity of electric charge q that passes any point in the circuit during the time Δt is represented by.
(1) $q = \frac{1}{R}.\frac{\Delta\phi}{\Delta t}$ (2) $q = \frac{\Delta\phi}{R}$
(3) $q = \frac{\Delta\phi}{\Delta t}$ (4) $q = R.\frac{\Delta\phi}{\Delta t}$

10. If a bar magnet is cut identically into two parts along AB as shown in figure.

Choose the **incorrect** option.
(1) Magnetic moment of each parts remains as before cutting
(2) Pole strength gets half after cutting
(3) Magnetic moment gets half after cutting for each part
(4) None of these

11. The $\chi - (1/T)$ graph for an alloy of paramagnetic nature is shown in figure. The curie constant is, then:



(1) 57 K
(2) 2.8×10^{-3} K
(3) 570 K
(4) 17.5×10^{-3} K

12. For a diamagnetic substance (symbols have their usual meanings)

(1) $\chi_m > 0, \mu_r > 1$
(2) $\chi_m < 0, \mu_r > 1$
(3) $\chi_m < 0, \mu_r < 1$
(4) $\chi_m > 0, \mu_r < 1$

SECTION-(II) CHEMISTRY

13. A student tried two reactions for preparing tert-butyl ethyl ether:

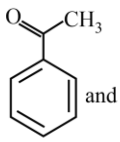
A. $C_2H_5ONa + CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-Cl \rightarrow$

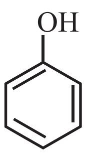
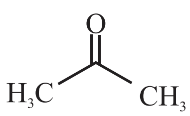
B. $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-ONa + CH_3-CH_2-Cl \rightarrow$

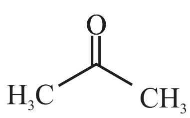
Which reaction will give tert-butyl ethyl ether?

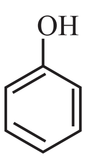
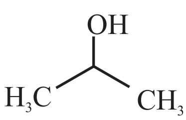
(1) A only
(2) B only
(3) A and B
(4) Neither A nor B

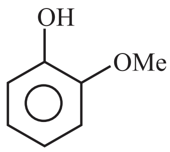
14. The products formed in the reaction of cumene with O_2 followed by treatment with dil. HCl are:

(1)  and CH_3 — OH

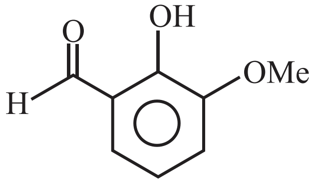
(2)  and 

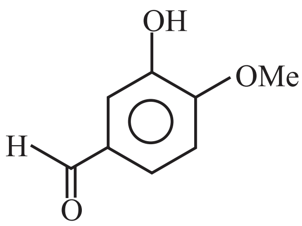
(3)  and 

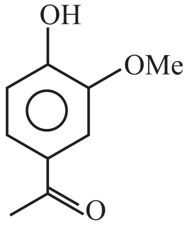
(4)  and 

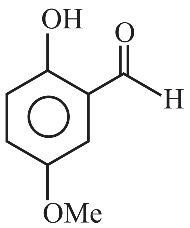
15.  $\xrightarrow[\text{(ii) } H_3O^{\oplus}]{\text{(i) KOH, } CHCl_3}$ 'P' (Major product)

Major product of the reaction is:

(1) 

(2) 

(3) 

(4) 

16. $CH_3CH_2Br + Mg \xrightarrow{\text{dry ether}} A \xrightarrow{H_2O} B$

The product B in the above reaction sequence is:

(1) CH_3CH_3
(2) $CH_2 = CH_2$
(3) $CH \equiv CH$
(4) CH_3CH_2OH

17. Given below are two statements:

Statement I: S_{N^1} reactions of optically active halides are accompanied by inversion of configuration.

Statement II: S_{N^2} reactions of optically active halides are accompanied by retention of configuration.

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

18. Aryl halides are less reactive towards nucleophilic substitution reactions than alkyl halides. This is due to:

(1) the formation of the less stable carbonium ion.
(2) resonance stabilization of halogen with ring.
(3) their longer carbon-halogen bond.
(4) the inductive effect.

19. The positive carbylamine test is given by:

(A) N, N-dimethyl aniline
(B) 2, 4-dimethyl aniline
(C) N-methyl-o-methyl aniline
(D) p-methyl benzyl amine

The **correct** option is:

(1) A and B (2) B and C
(3) A and C (4) B and D

20. Aniline on treatment with bromine water yields white precipitate of:

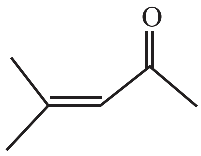
(1) o–Bromoaniline
(2) p–Bromoaniline
(3) 2, 4, 6–Tribromoaniline
(4) m–Bromoaniline

21. In the diazotisation of aryl amine, the use of nitrous acid is:

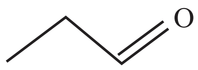
(1) it suppresses hydrolysis of phenol.
(2) it is a source of electrophilic nitrosonium ion.
(3) it neutralizes the base liberated.
(4) all of these

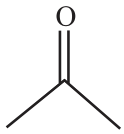
22. The **correct** sequence of reaction to be performed to convert benzene into m-bromoaniline is:

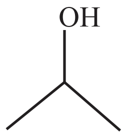
(1) Nitration, reduction, bromination
(2) Bromination, nitration, reduction
(3) Nitration, bromination, reduction
(4) Reduction, nitration, bromination

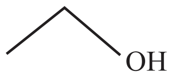
23. (x) $\xrightarrow[\Delta]{\text{dil.KOH}}$ 

Structure of (x) is:

(1) 

(2) 

(3) 

(4) 

24. Which reagent will perform the following reduction?

$$CH_3 - CH = CH - CHO \rightarrow CH_3 - CH = CH - CH_2OH$$

(1) $LiAlH_4$
(2) $NaBH_4$
(3) H_2/Ni
(4) $Na/Hg + H_2O$

[Short Practice Test-07 | Yakeen Plus (2025) | 30/10/2024]

Photosynthesis in Higher Plants, Plant Growth and Development, Sexual Reproduction in Flowering Plant

SECTION-(III) BOTANY

25. Select the **mismatched** pair from the following.

(1)	IAA	Indole compound
(2)	GA	Terpenes
(3)	ABA	Carotenoid derivatives
(4)	Ethylene	Adenine derivatives

26. Consider the following statements w.r.t gibberellins (GAs).

A. More than 100 GAs are reported from fungi only.
B. All GAs are acidic.
C. These help in early seed production in conifers.
D. They also delay senescence.

Choose the most appropriate answer from the options given below:

(1) D is correct only
(2) A, B and D are correct only
(3) B, C and D are correct only
(4) A and B are correct only

27. Read the following statements and select the **incorrect** option.

(1) Photosystem-I receives electrons from photosystem-II.

(2) Photosystem-II receives electrons from photolytic dissociation of water.

(3) Formation of NADPH is associated with photosystem-II.

(4) Reaction centre of photosystem I is P700.

28. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Most zygotes divide only after certain amount of endosperm is formed.

Reason R: Endosperm formation is an adaptation to provide assured nutrition to the developing embryo.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is not the correct explanation of A.

29. In arithmetic growth rate, when length of the organ is plotted against time, the nature of graph curve will be:

(1) linear. (2) sigmoidal.

(3) parabolic. (4) hyperbolic.

30. The ability of plant to follow different pathways and produce different structures in response to environment and phases of life is termed as:

(1) differentiation. (2) growth efficiency.

(3) plasticity. (4) heterophylly.

31. Given below are two statements:

Statement I: The microsporangia is surrounded by four wall layers.

Statement II: Each lobe of anther is dithecous.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

32. Read the following statements regarding pollen.

A. A large number of pollen products in the form of tablets and syrups are available in the market.

B. Pollens have fascinating array of patterns and designs of pectin on outer wall.

C. Pollens can be used as food supplement.

D. Pollen grains are rich in nutrients.

Choose the most appropriate answer from the options given below:

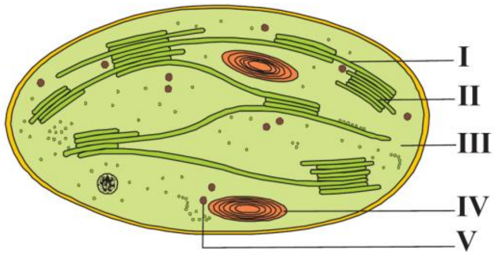
(1) A, C and D are correct only

(2) A and C are correct only

(3) B, C and D are incorrect

(4) D is correct only

33. The given figure shows the diagrammatic representation of a section of chloroplast. Few structures are marked as **I**, **II**, **III**, **IV & V**.



Combination of which parts is responsible for trapping the light energy and utilisation of ATP and NADPH?

(1) I, II and III

(2) II, III and IV

(3) III and IV

(4) I, IV and V

34. Match **List-I** with **List-II:**

List-I		List-II	
(A)	*Vallisneria*	(I)	Wind pollination
(B)	Seagrasses	(II)	Provide safe place to lay eggs for insect
(C)	Corn	(III)	Female flowers reach the surface of water by the long stalk
(D)	*Amorphophallus*	(IV)	Female flowers remain submerged

Choose the **correct** answer from the option given below:

(1) A-III, B-II, C-I, D-IV

(2) A-I, B-III, C-II, D-IV

(3) A-IV, B-I, C-III, D-II

(4) A-III, B-IV, C-I, D-II

35. In C_4 plants the saturation point of CO_2 is:
(1) beyond 450 μlL^{-1}.
(2) beyond 540 μlL^{-1}.
(3) at about 630 μlL^{-1}.
(4) at about 360 μlL^{-1}.

36. Calvin cycle requires __________ during carboxylation.
(1) NADPH + H^+ and CO_2
(2) water and ribulose bisphosphate only
(3) ribulose bisphosphate, water and CO_2
(4) 3PGA and CO_2

SECTION-(IV) ZOOLOGY

37. Read the following statements (**I-V**).
I. After ovulation, remaining part of Graafian follicle transforms into corpus luteum.
II. The secondary oocyte forms a membrane around itself called zona pellucida.
III. The primary oocyte within the tertiary follicle completes its first meiotic division.
IV. No menstruation occurs during pregnancy.
V. Sperm formation stops in old men.
Which of the above statements are **correct**?
(1) I, II and III only
(2) I, III and IV only
(3) I, II, III and IV only
(4) I, II, III, IV and V

38. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Parturition	(I)	Cessation of menstrual cycle
(B)	Implantation	(II)	Attachment of blastocyst to endometrium
(C)	Menopause	(III)	Delivery of baby
(D)	Birth canal	(IV)	Cervical canal alongwith vagina

Choose the **correct** answer from the options given below:
(1) A-III, B-II, C-I, D-IV
(2) A-I, B-III, C-II, D-IV
(3) A-II, B-I, C-III, D-IV
(4) A-III, B-I, C-II, D-IV

39. Arrange the following events of human female reproductive cycle in **correct** sequence.
I. Secretion of FSH
II. Growth of corpus luteum
III. Growth of follicles
IV. Ovulation
V. Sudden increase in level of LH
(1) III → I → IV → II → V
(2) I → III → V → IV → II
(3) I → IV → III → V → II
(4) II→ I → III → IV → V

40. Which of the following is an unpaired structure in male reproductive system?
(1) Bulbourethral gland
(2) Seminal vesicle
(3) Prostate gland
(4) Ureters

41. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Rate of the chemical reaction doubles on increasing the temperature by 10°C.
Reason R: Catalysed reactions proceed at rates higher than that of uncatalysed reactions.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

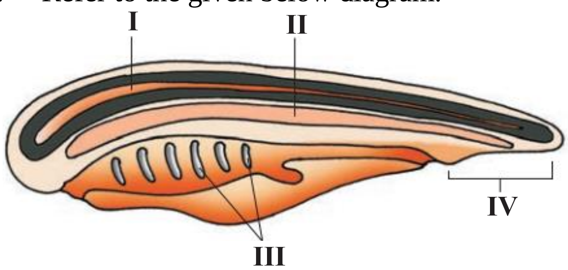
42. Which of the following is **not** correct with respect to enzymes?
(1) Enzymes show maximum activity at an optimum pH and temperature.
(2) Enzymes get denatured at high temperature.
(3) Enzymes are highly specific in nature.
(4) Most enzymes are proteins but some are carbohydrates in nature.

43. Given below are two statements:
Statement I: Cellulose cannot hold I_2.
Statement II: Exoskeletons of arthropods have a simple polysaccharide called chitin.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

44. Choose the **incorrect** statement.
(1) Scent is a secondary metabolite.
(2) GLUT-4 enables glucose transport into cells.
(3) Ricin is an alkaloid.
(4) Complex polysaccharides are mostly homopolymers.

45. Select the **incorrect** option.
(1) *Taenia*: Tapeworm
(2) *Meandrina*: Brain coral
(3) *Wuchereria*: Hookworm
(4) *Hirudinaria*: Blood-sucking leech

46. Refer to the given below diagram.



Which of the following represents nerve cord?
(1) I (2) II
(3) III (4) IV

47. Select the **incorrect** statement.
(1) Protochordates are exclusively marine.
(2) In urochordates, notochord extends from head to tail.
(3) In *Antedon*, the excretory system is absent.
(4) Cyclostomes are ectoparasites on some fishes.

48. The following characteristics are **correct** for which organism's class?
A. Includes both marine and freshwater organisms.
B. Bony endoskeleton.
C. Air bladder regulates buoyancy.
(1) *Labeo*
(2) *Trygon*
(3) *Scoliodon*
(4) Both (1) and (3)